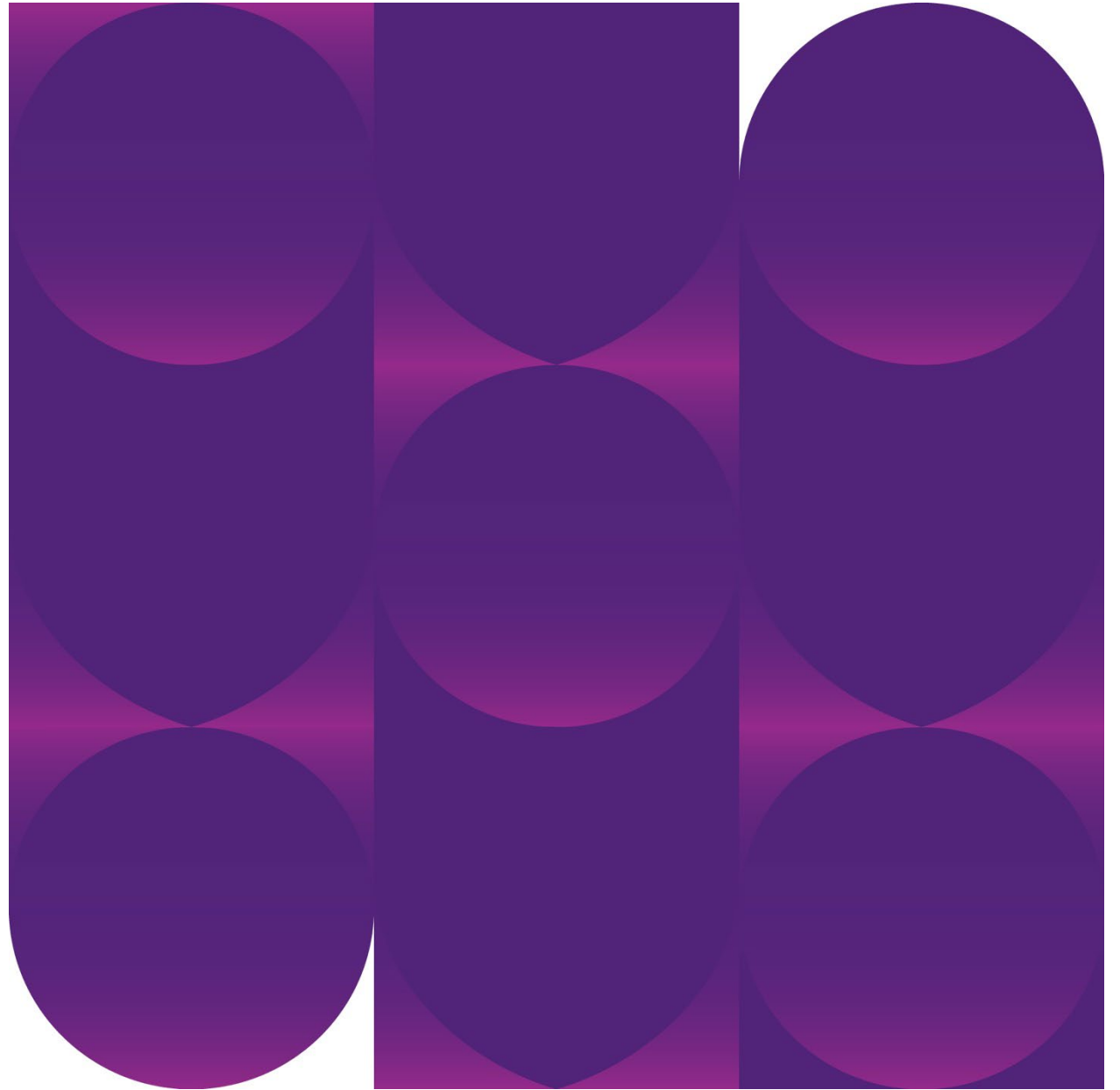


Embryology of head, face and oral cavity (Part 1)

Dr Sowmya Shetty

s.shetty@uq.edu.au

DENT1020 Dental Science I



Cell biology summary

1. An appreciation of the basic structures and functions of eukaryotic (and prokaryotic) cells can inform many aspects of dentistry, e.g. antibiotic usage
2. Cells are the basic structural and functional units of every organism
3. They are of two distinct types: prokaryotic and eukaryotic
4. The nucleus contains most of the genes in the eukaryotic cells.
5. Prokaryotes undergo cell division by binary fission
6. Eukaryotes undergo cell division by the cell cycle
7. The cell cycle consists of four phases – G1, S, G2 and M
8. Asexual reproduction occurs by mitosis
9. In sexual reproduction, two haploid gametes fuse to form a zygote
10. Haploid gametes are produced by the process of meiosis

This lecture...

By the end of this lecture, you should be able to:

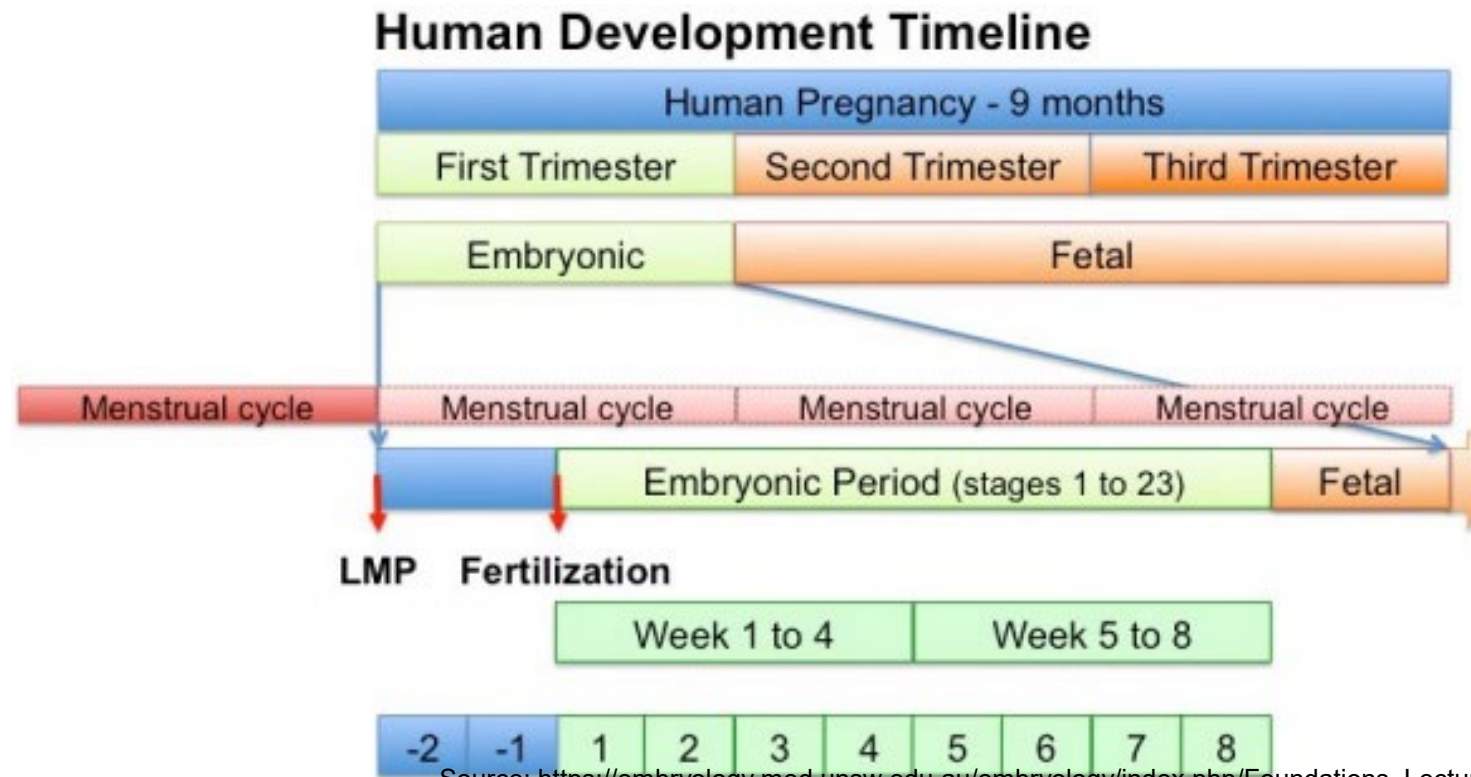
- Describe the general stages of human development from fertilisation through to birth
- Describe which types of tissues arise from the different germ layers
- Compare and contrast the differences between embryonic and foetal stages of development
- Describe the developmental origin of orofacial structures, i.e., tissues derived from the pharyngeal arches

Suggested additional reading:

- *Nanci A. Ten Cate's Oral Histology: Development, Structure and Function. 9th edn. St Louis, Missouri: Elsevier. Chapter 2 & 3.*
- *Fehrenbach MJ & Popowics T. Illustrated Dental Embryology, Histology, and Anatomy. 5th edn. Missouri, USA. Elsevier Saunders. 2020. Chapter 3 & 4*

Embryonic and fetal stages

- During weeks 4-8, the body's various tissues, organs and limbs begin to develop
- This is known as **organogenesis**, and occurs during the **1st trimester** of pregnancy
- The **fetal growth** period occurs during the **2nd and 3rd trimesters**



Sequences of prenatal development

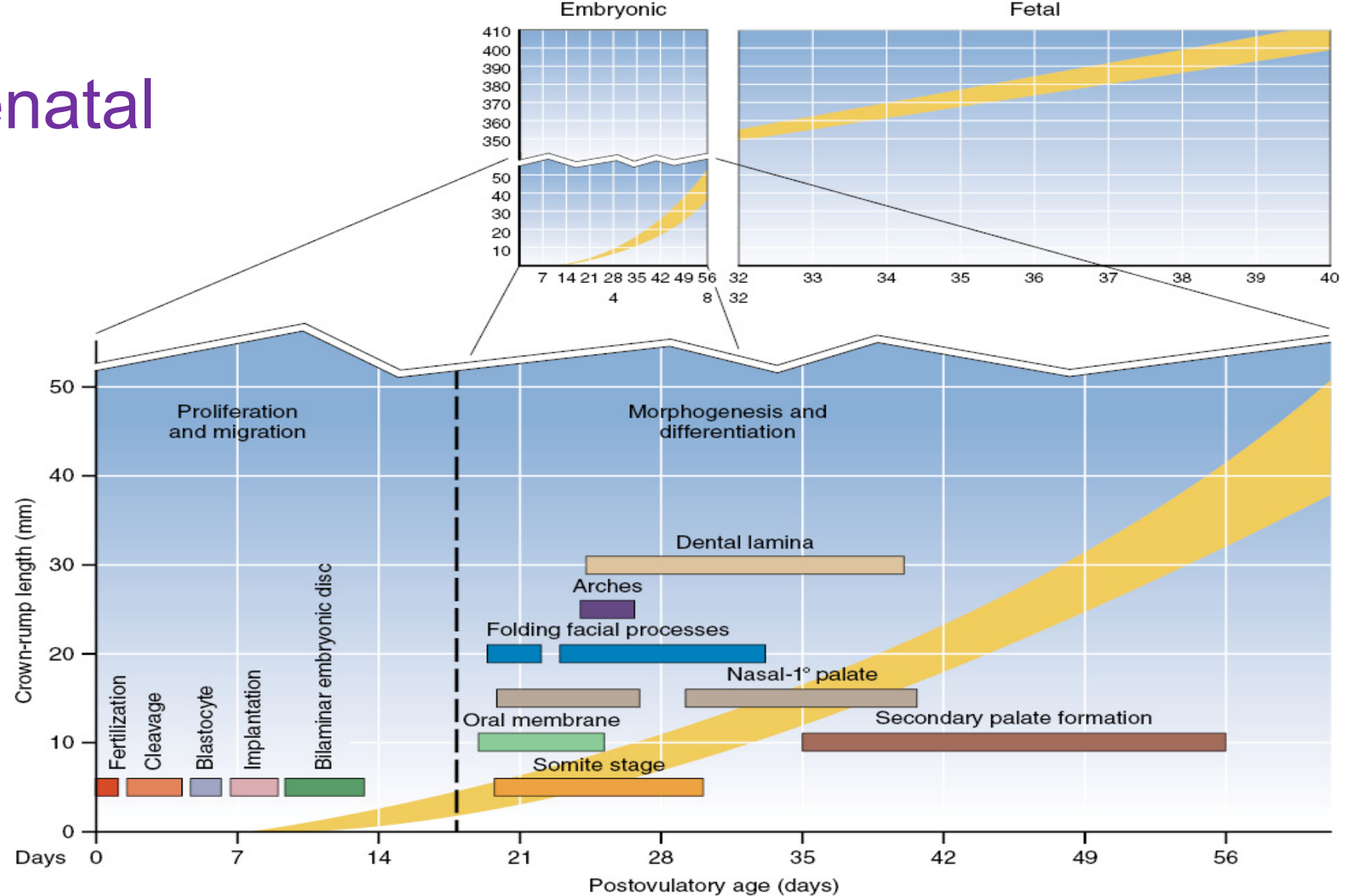


FIGURE 2-2 Sequences of prenatal development. The upper diagram shows the distinction between embryonic and fetal stages. The lower part of the embryonic diagram is expanded in the bottom diagram, which distinguishes the stages of proliferation and migration and morphogenesis and differentiation. The timing of key events also is indicated.

1. Before fertilisation

- **Meiosis** in gonads produces **haploid gametes**
- Beginning at puberty, females release one egg during each menstrual cycle
- Males produce around two thousand sperm every second

2. Fertilisation and implantation

- **Gametes** fuse in the fallopian tube to create a **zygote** (fertilisation)
- **Cleavage** (mitosis) continues to form a multicellular **morula** (“mulberry” shaped)

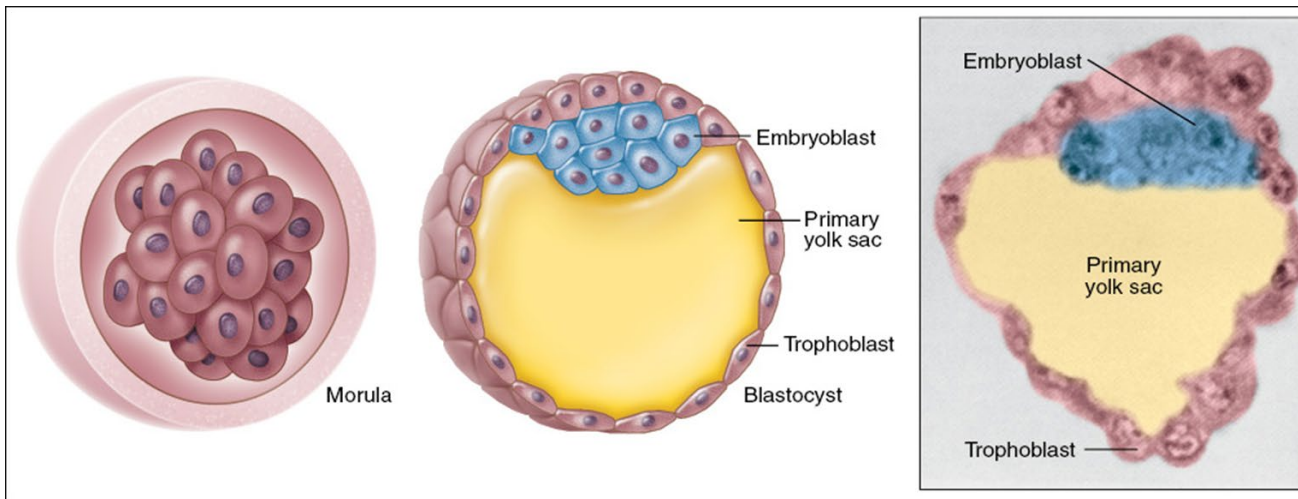


FIGURE 2-5 Differentiation of the morula into a blastocyst. At this time cells differentiate into the embryoblast (involved in development of the embryo) and the trophoblast (involved in maintenance).

(Adapted from Hertig AT et al: *Contrib Embryol* 35:199, 1954.)

This progresses to a **blastocyst**, which has an **inner cell mass (embryoblast)** that forms the embryo and an **outer cell mass (trophoblast)** which forms the placenta

Implantation of the blastocyst occurs in the uterus (womb) around 7-10 days later

3. Bilaminar germ disc

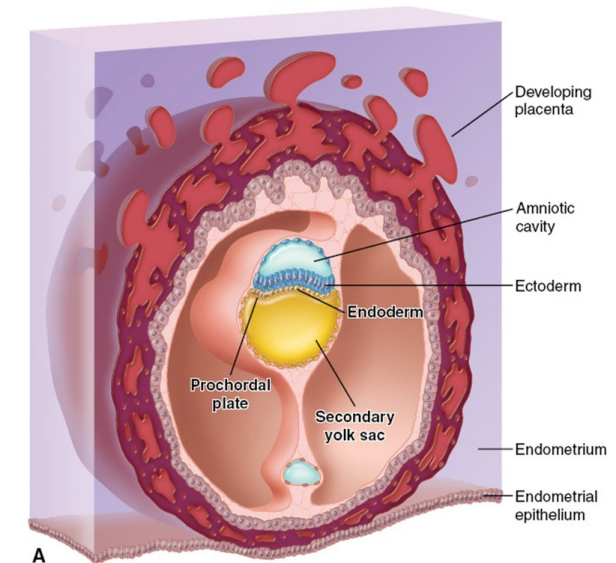
Approximately day 8 of gestation, the cells of the **embryoblast** differentiate to form a two-layered disk called the **bilaminar germ disc**

The cells situated **dorsally**, or the ectodermal layer, are columnar and reorganise to form the amniotic cavity.

The cells situated **ventrally**, the endodermal layer, are cuboidal and form the roof of a second cavity (the secondary yolk sac).

This configuration is completed after 2 weeks of development.

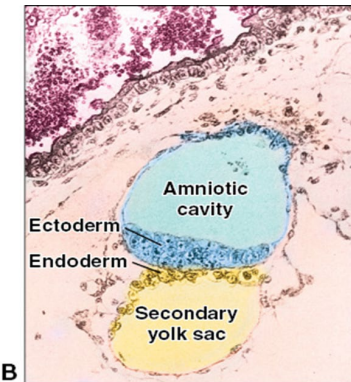
During this time the axis of the embryo is established and is represented by a slight enlargement of the ectodermal and endodermal cells at the head (**cephalic or rostral**) end of the embryo in a region known as the **prochordal (or prechordal) plate**, where ectoderm and endoderm are in contact.



A

FIGURE 2-6A A, Schematic representation and B, histologic section of a human blastocyst at 13 days. An amniotic cavity has formed within the ectodermal layer. Proliferation of endodermal cells forms a secondary yolk sac. The bilaminar embryo is well established.

Copyright © 2013 by Mosby, an imprint of Elsevier Inc.



B

FIGURE 2-6B A, Schematic representation and B, histologic section of a human blastocyst at 13 days. An amniotic cavity has formed within the ectodermal layer. Proliferation of endodermal cells forms a secondary yolk sac. The bilaminar embryo is well established.

(Adapted from Brewer JI: *Contrib Embryol* 27:85, 1938.)

3. Bilaminar germ disc

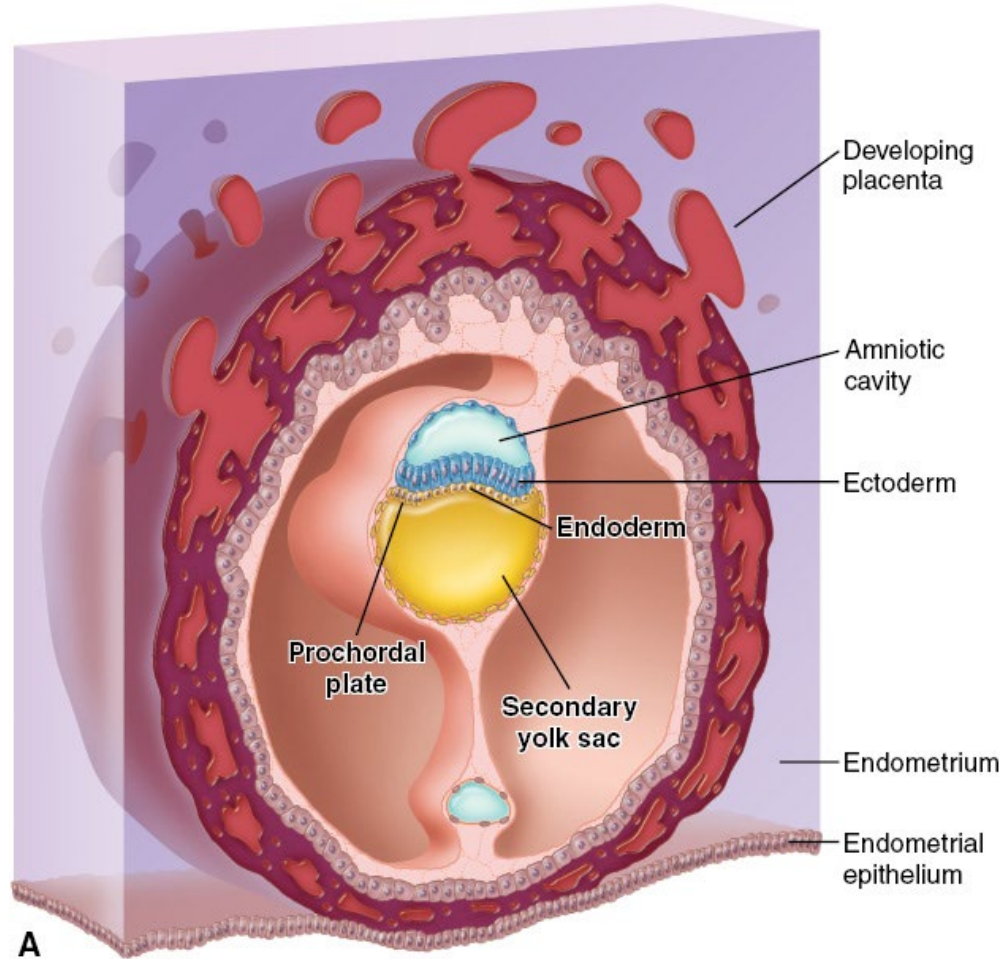


FIGURE 2-6A **A**, Schematic representation and **B**, histologic section of a human blastocyst at 13 days. An amniotic cavity has formed within the ectodermal layer. Proliferation of endodermal cells forms a secondary yolk sac. The bilaminar embryo is well established.

Copyright © 2013 by Mosby, an imprint of Elsevier Inc.

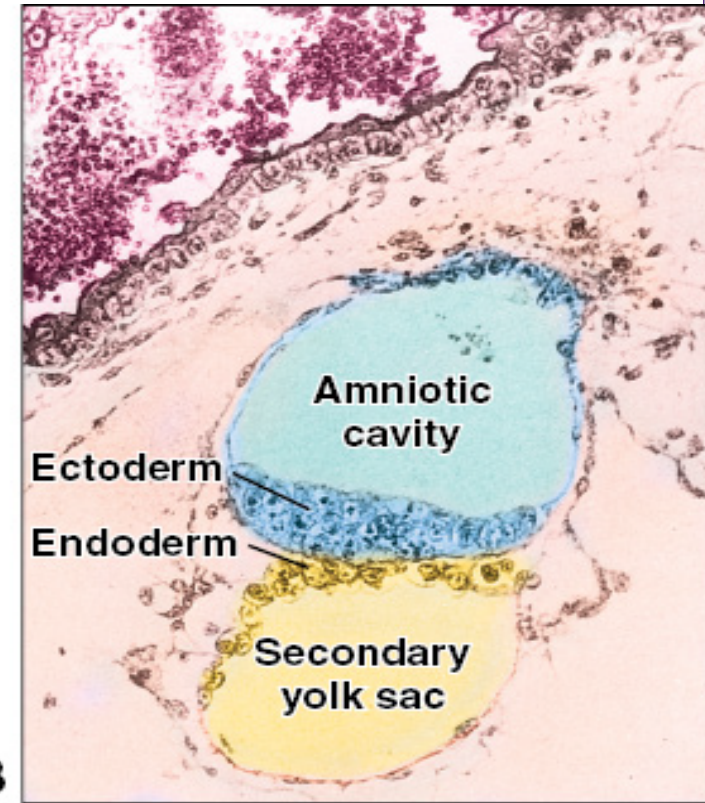


FIGURE 2-6B **A**, Schematic representation and **B**, histologic section of a human blastocyst at 13 days. An amniotic cavity has formed within the ectodermal layer. Proliferation of endodermal cells forms a secondary yolk sac. The bilaminar embryo is well established.

(Adapted from Brewer JI: *Contrib Embryol* 27:85, 1938.)

From Nanci A. Ten Cate's *Oral Histology: Development, Structure and Function*. 9th edn. St Louis, Missouri: Elsevier, 2018, p15

Recap: Sequences of prenatal development

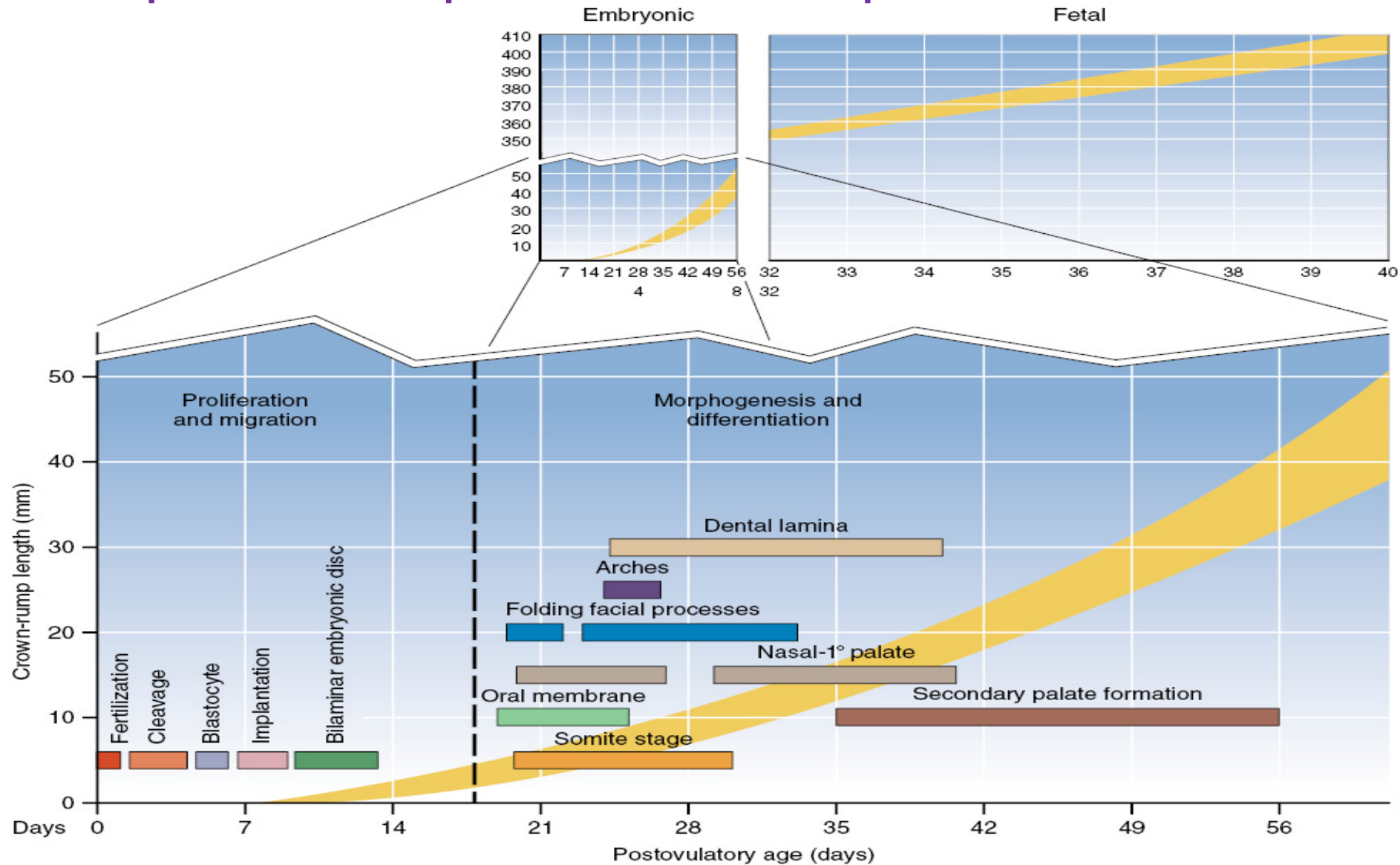


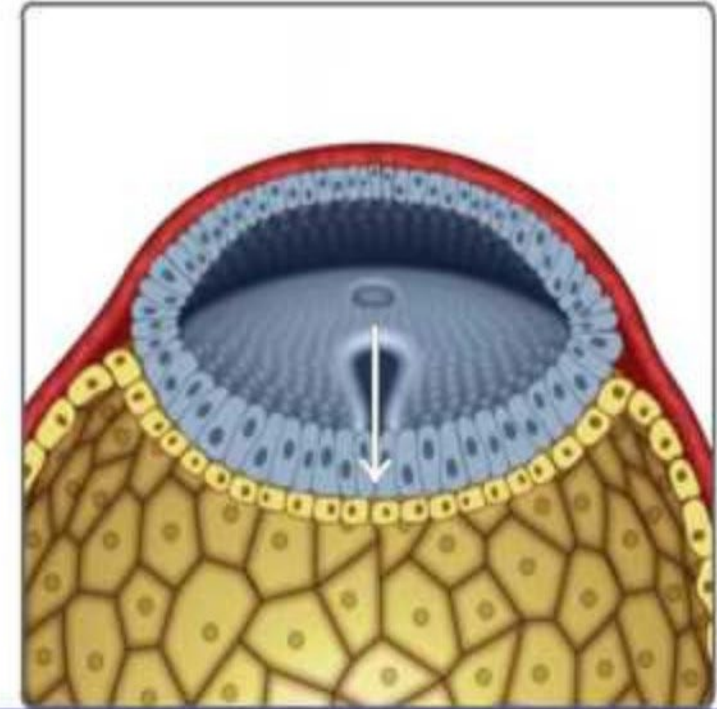
FIGURE 2-2 Sequences of prenatal development. The upper diagram shows the distinction between embryonic and fetal stages. The lower part of the embryonic diagram is expanded in the bottom diagram, which distinguishes the stages of proliferation and migration and morphogenesis and differentiation. The timing of key events also is indicated.

From Nanci A. Ten Cate's *Oral Histology: Development, Structure and Function*. 9th edn. St Louis, Missouri: Elsevier, 2018, p13

4. Gastrulation

A phase early in embryonic development (around week 3) during which the single-layered **blastula** reorganises into a **three-layered (trilaminar)** structure known as the **gastrula**.

These **germ layers** are the **ectoderm**, **mesoderm** and **endoderm**

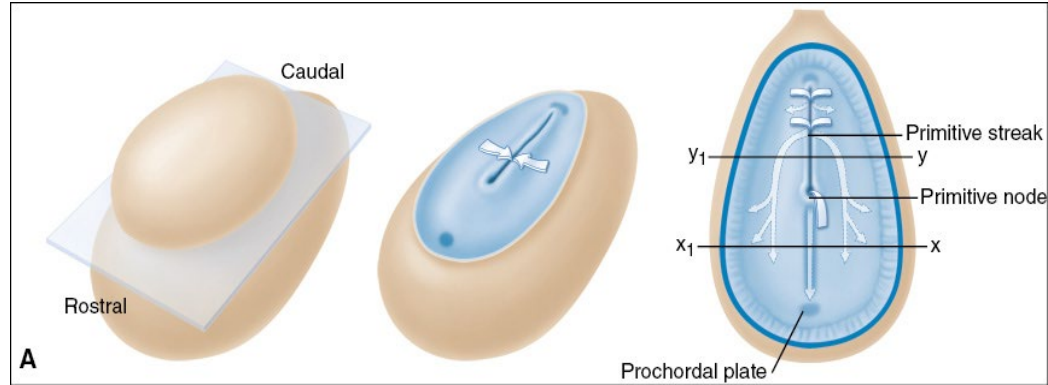


This depression continues along the midline of the epiblast, towards the caudal end of the streak.

<https://www.youtube.com/watch?v=ADlYn0ImTNg>

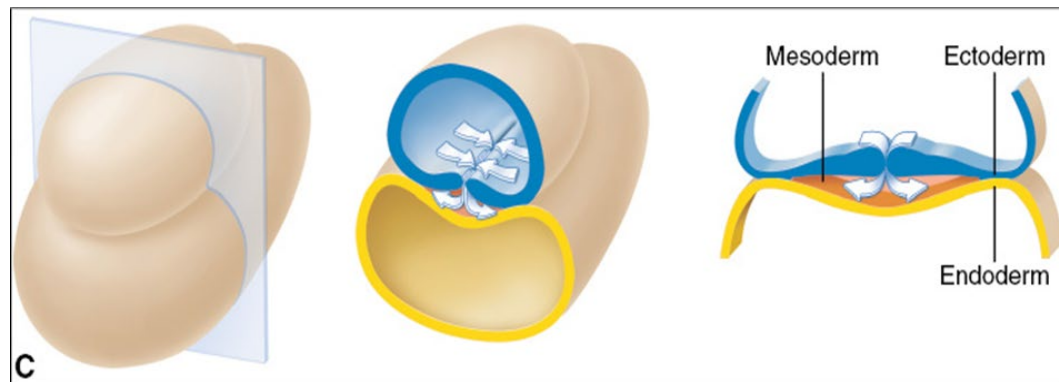
- The floor of the amniotic cavity is formed by ectoderm, and within it a structure called the primitive streak develops along the midline by cellular convergence.
- The rostral end of the streak finishes in a small depression called the primitive node or primitive pit.
- Cells of the ectoderm migrate through the streak and between the ectoderm and endoderm. The notochord is formed to support the primitive embryo.
- Primitive streak invaginates and tube of ectodermal cells forms inside the mesoderm

4. Gastrulation Conversion of the bilaminar embryo into a trilaminar embryo.

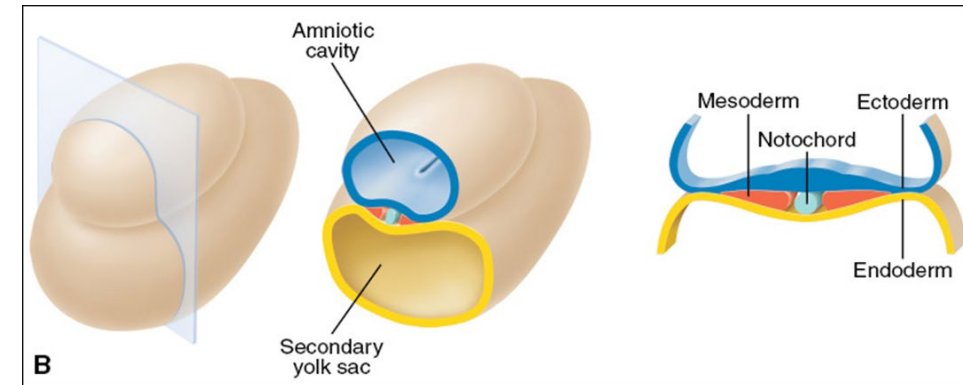


A: Floor of the amniotic cavity formed by the ectodermal layer of the bilaminar embryo. Ectodermal cells converge towards midline to form **primitive streak** (a narrow groove terminating in a circular depression called primitive node).

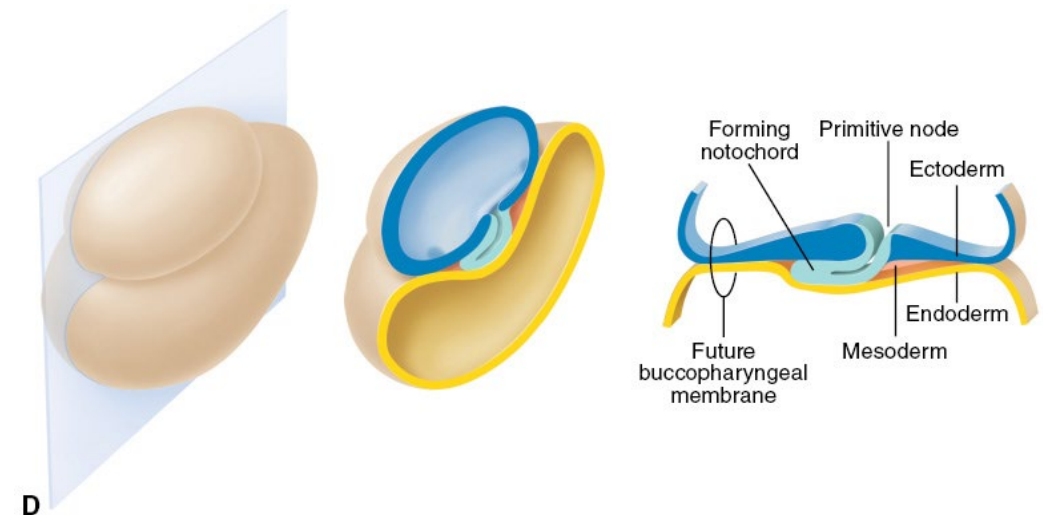
Ectodermal cells migrate through streak and between in ectodermal and endodermal layers (in lateral and cephalic directions).



C: Section through y-y1



B: A notochord process extends forward from the primitive node. A transverse section through x-x1 showing notochord flanked by mesoderm.



D: Notochord pushing rostrally as seen in longitudinal section.

5. Neurulation

Formation of the neural tube, which goes on to form the nervous system

- The folding process in vertebrate embryos and segmentation of the ectoderm to form the **neural tube**
- The nervous system develops as a thickening within the ectodermal layer at the rostral end of the embryo. This thickening constitutes the neural plate, which rapidly forms raised margins (neural folds).
- These folds encompass and delineate a deepening midline depression, the neural groove. The neural folds eventually fuse so that a neural tube separates from the ectoderm to form the floor of the amniotic cavity, with mesoderm intervening.
- <https://www.youtube.com/watch?v=IGLexQR9xGs>

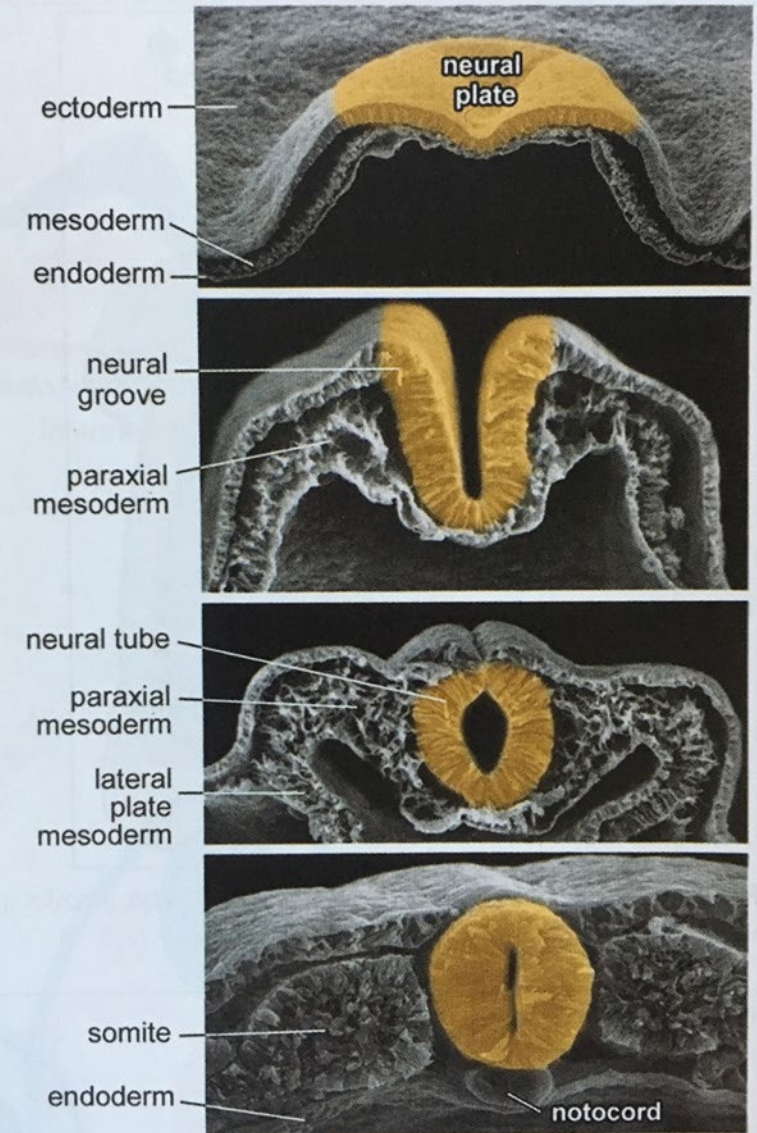
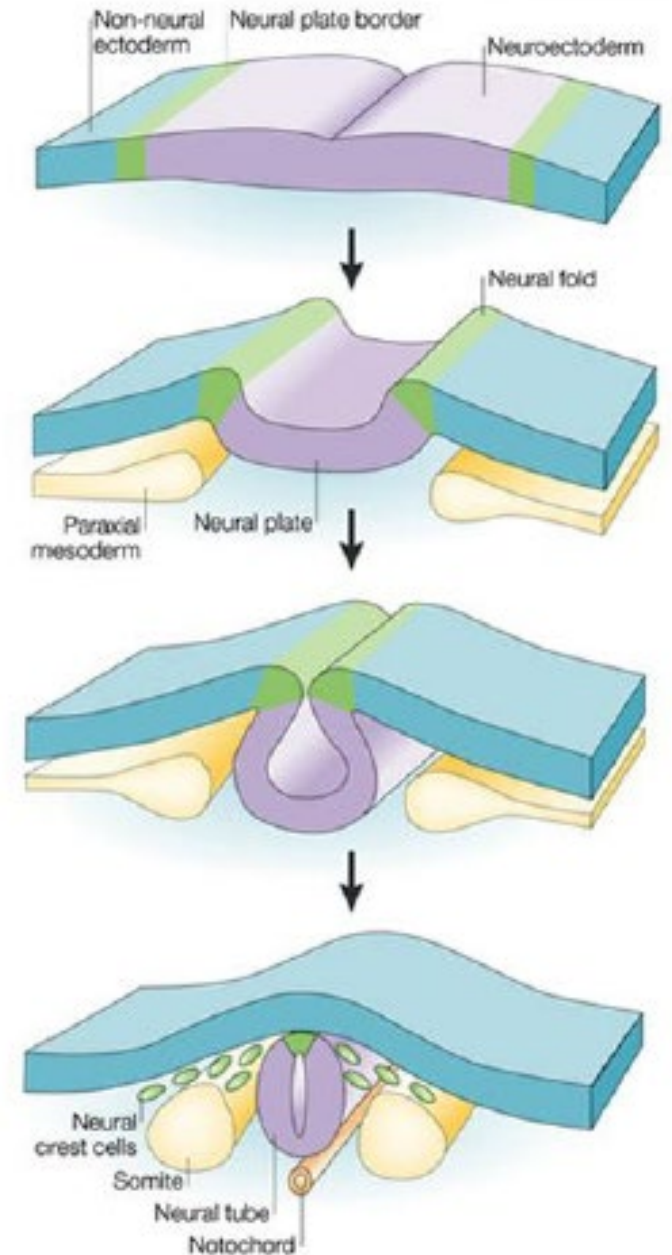


FIGURE 2-8 Scanning electron micrograph views of formation and closure of the neural fold elevations. (Courtesy G. Schoenwolf).

5. Neurulation

- Neural crest cells break off/develop from this neural tube.
- These neural crest cells make most of your face. The ectoderm lines your mouth (oral epithelium). All of your tooth, apart from ameloblasts, are derived from these neural crest cells (ameloblasts are derived from ectoderm).
- In the rest of the body, it's the mesoderm that goes on to make most of the structures.



Recap: Sequences of prenatal development

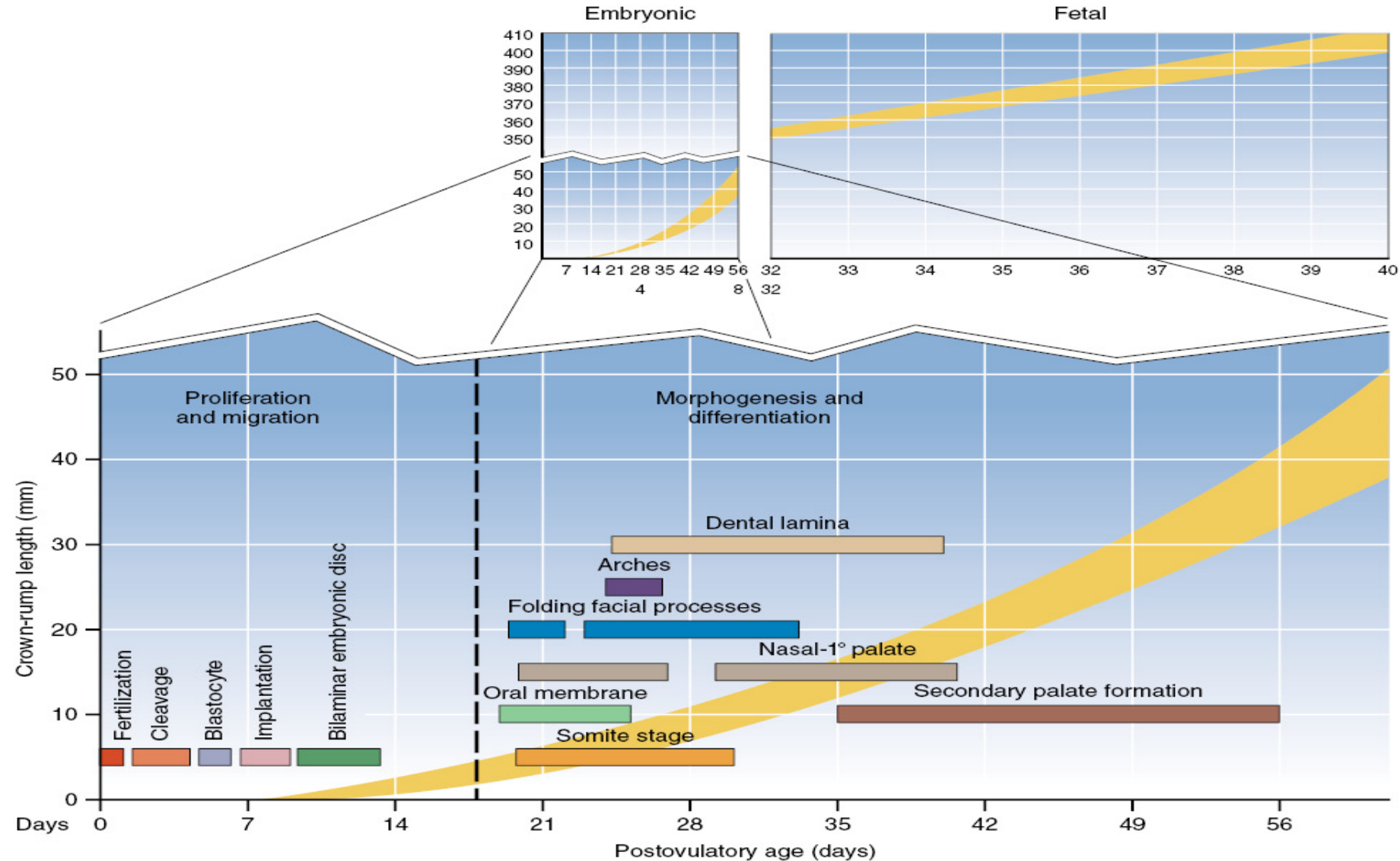
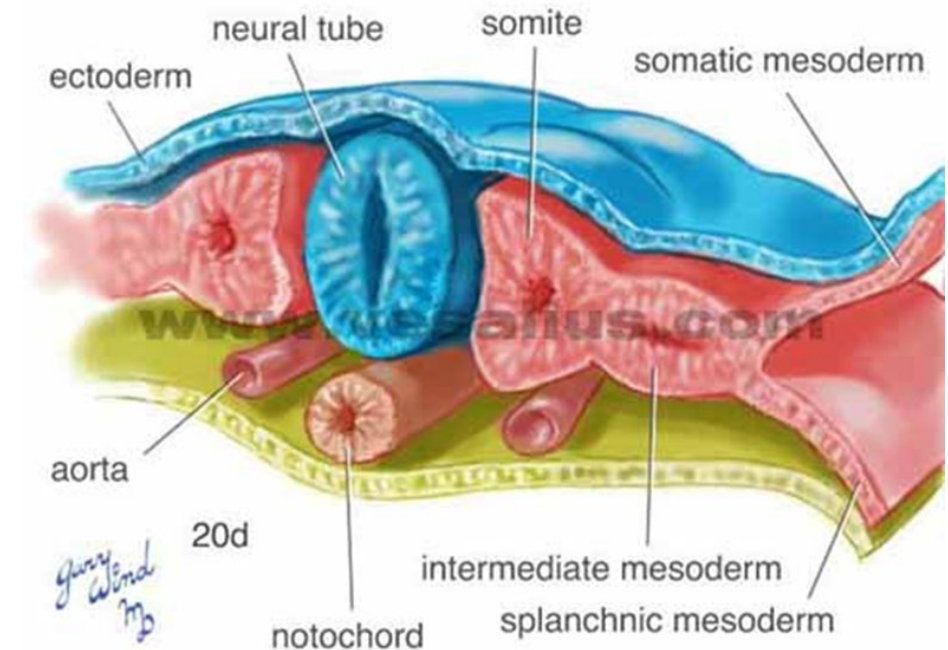


FIGURE 2-2 Sequences of prenatal development. The upper diagram shows the distinction between embryonic and fetal stages. The lower part of the embryonic diagram is expanded in the bottom diagram, which distinguishes the stages of proliferation and migration and morphogenesis and differentiation. The timing of key events also is indicated.

6. Somitogenesis

- As the neural tube forms, changes occur in the mesoderm adjacent to the tube and the notochord.
- The mesoderm first thickens on each side of the midline to form **paraxial mesoderm**. Along the trunk (anterior-posterior axis) of the embryo, this paraxial mesoderm breaks into segmented blocks called **somites**.
- Each **somite** has three components: (1) the **sclerotome**, which eventually contributes to two adjacent vertebrae and their discs; (2) the **myotome**, which gives origin to a segmented mass of muscle; and (3) the **dermatome**, which gives rise to the connective tissue of the skin overlying the **somite**.

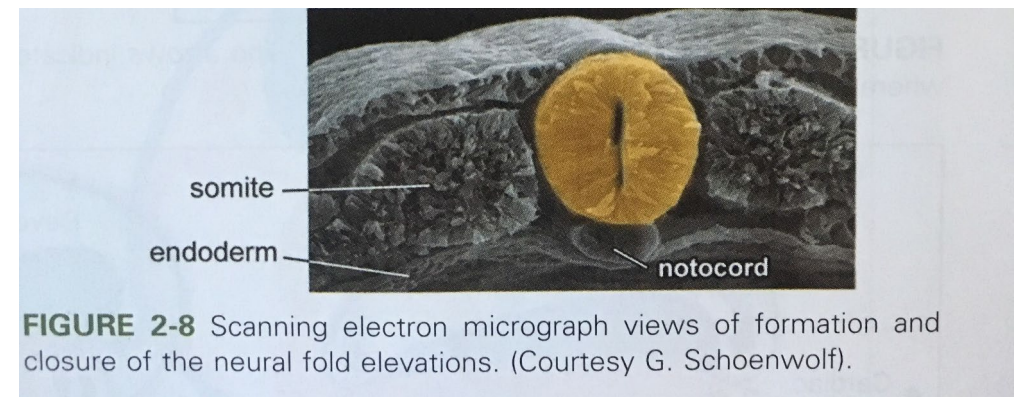
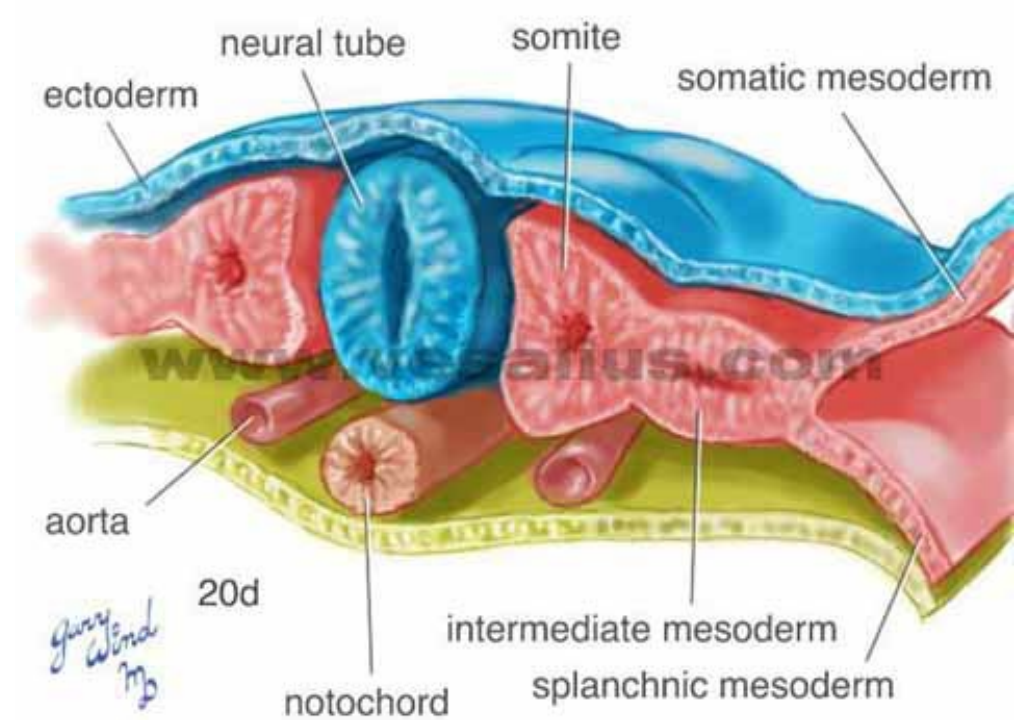


6. Somitogenesis

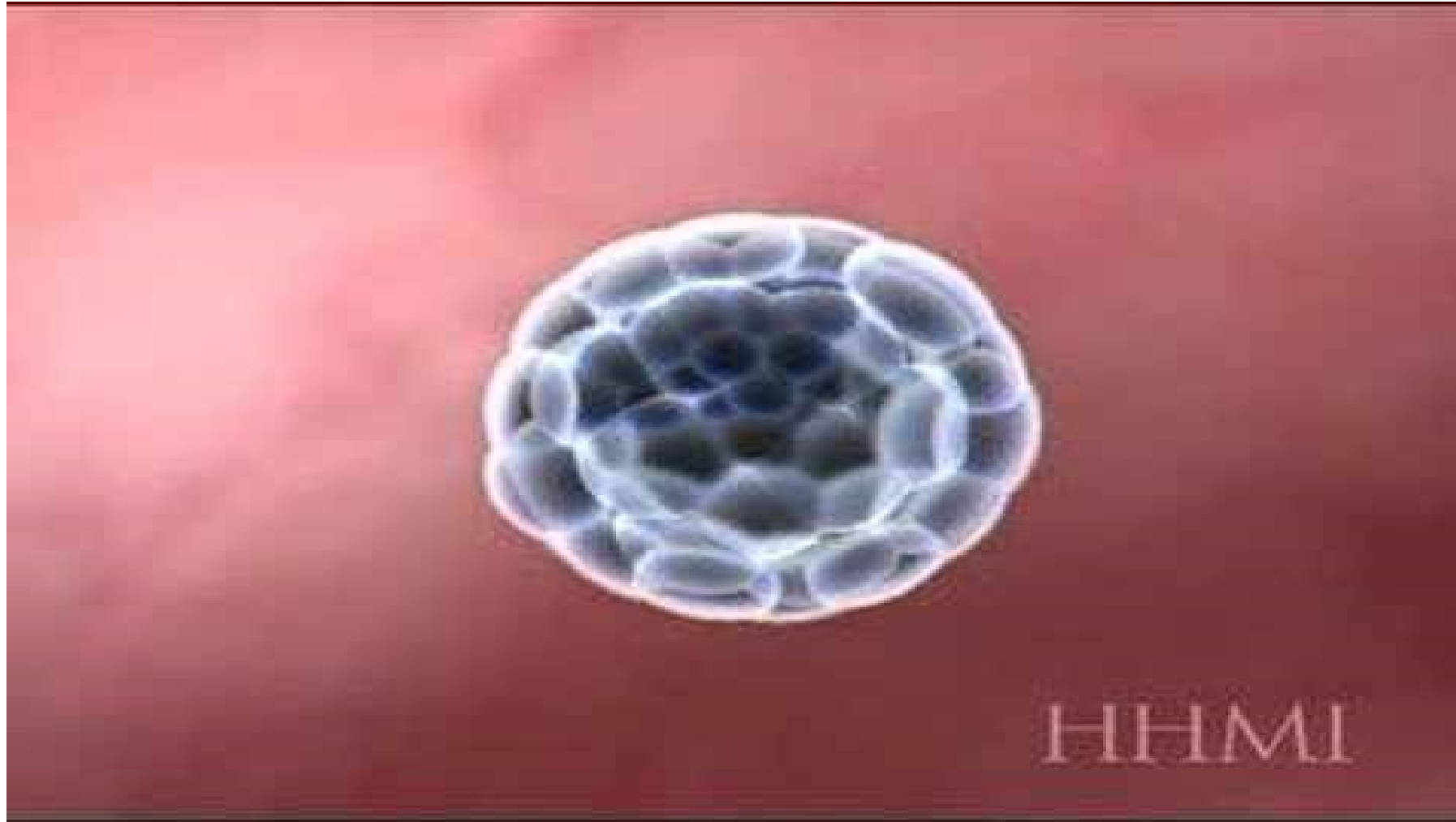
Somites give you segmentation in the way the body develops.

In vertebrates, they give rise to skeletal muscle, cartilage, tendons, endothelial cells, and dermis

Each muscle/organ comes from a predetermined embryonic region, it's not random



Video: Human Development



<https://youtu.be/UgT5rUQ9EmQ>

Sequences of prenatal development

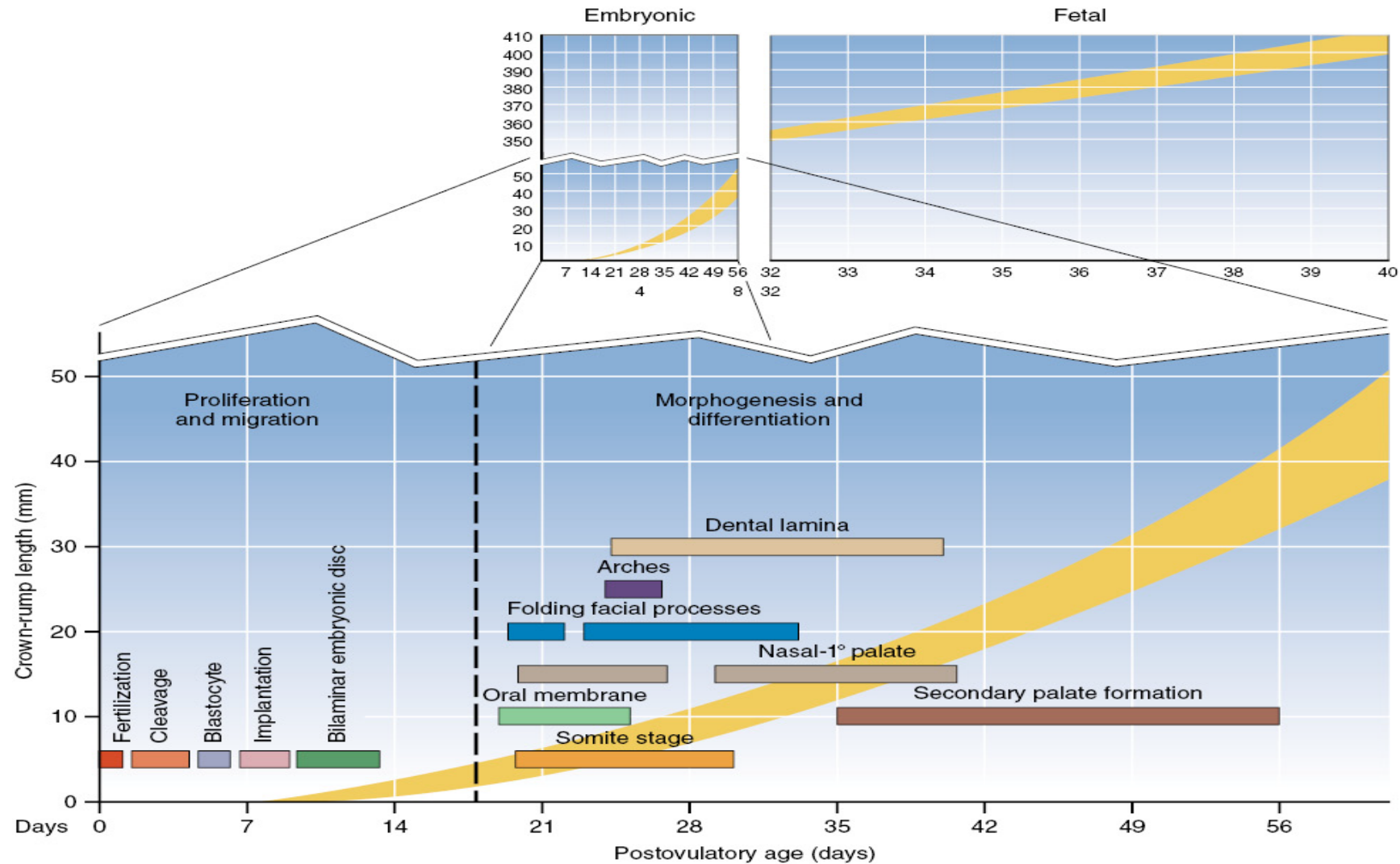


FIGURE 2-2 Sequences of prenatal development. The upper diagram shows the distinction between embryonic and fetal stages. The lower part of the embryonic diagram is expanded in the bottom diagram, which distinguishes the stages of proliferation and migration and morphogenesis and differentiation. The timing of key events also is indicated.

Summary

Sequences of prenatal development:

- Before fertilisation meiosis in gonads produces haploid gametes
- Gametes fuse in the fallopian tube to create a zygote (fertilisation)
- Cleavage (mitosis) continues to form a multicellular morula
- This progresses to a blastocyst, which has an inner cell mass (embryoblast) that forms the embryo and an outer cell mass (trophoblast) which forms the placenta
- Implantation of the blastocyst occurs in the uterus (womb) around 7-10 days later
- The embryoblast differentiates to form the bilaminar germ disc
- Gastrulation occurs and a trilaminar germ disc forms
- Neurulation follows, the neural tube forming the nervous system
- Somitogenesis occurs, giving segmentation in the way the body develops

Questions?





THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Thank you

Dr Sowmya Shetty
DENT1020 course coordinator
s.shetty@uq.edu.au

Slides adapted from
Drs Bartle, Hosseinpour and Fernandez
(2018-2025)